St. Francis Institute of Technology, Mumbai-400 103
Department Of Information Technology

A.Y. 2025-2026
Class: BE-ITA/B, Semester: VII
Subject: Healthcare Lab

Experiment 7

1. **Aim:** To analyze data extracted from Twitter data during COVID .

2. **Objectives:** Students should be able to apply appropriate NLP techniques to analyze data extracted from social media.
3. **Prerequisite:** Python basics, A basic understanding of the goals and terminology of Natural Language Processing
4. **Pre-Experiment Exercise:**

Theory:

Natural Language Processing

Natural language processing (commonly referred to as NLP) is a subset of Artificial Intelligence research, which is concerned with machine learning modeling tasks, aimed at giving computer programs the ability to understand human language, both written and spoken.

Social Media Platform (Twitter)

Twitter is a platform where most of the people express their feelings towards the current context. As humans, we can guess the sentiment of a sentence whether it is positive or negative.
During the crisis situation caused due to COVID-19 disease, managing mental health and psychological well-being is as important as physical health of people. As social media is widely used by people to express their emotion and opinion, Twitter data posted by people during this crisis situation was the best source to analyze the emotions of people. For processing the cleaned data, use the NRC Word-Emotion Association Lexicon (have aka EmoLex). NRC Word-Emotion Association Lexicon is a list of English with real-valued scores of intensity for eight basic emotion words ns (anger, anticipation, disgust, fear, joy, sadness, surprise, and trust). Classify the text content of a tweeter dataset created by fetching tweets across the world into basic emotions like anger, anticipation, disgust, fear, joy, sadness, surprise and trust. This analysis can be used by authorities to understand the mental health of the people and can take necessary measures to decide on policies to fight against coronavirus which is affecting the social well-being as well as economy of the whole world.

Covid Dataset to perform Twitter Analysis :

Columns (Features):

- Location- Location of user

- Tweet At- Date of a tweet

- Original Tweet- actual tweet text

- Sentiment- sentiments(we can say emotions) like positive, negative, neutral, etc

5. Laboratory Exercise:

Steps of Extraction of Tweets using Tweepy:

- **Create a Twitter Developer Account**: To access Twitter's API, you will need to create a Twitter Developer Account and apply for developer access.
- **Obtain API Credentials**: After you have been approved for developer access, you will need to obtain your API credentials, including consumer key, consumer secret, access token, and access token secret.
- Install Tweepy: Use the pip command to install Tweepy in your Python environment.
- **Import Tweepy and Authenticate**: Import the Tweepy library into your Python script and authenticate your API credentials using the OAuthHandler and set_access_token methods.
- **Extract Tweets**: Use Tweepy functions like search or user_timeline to extract tweets based on keywords, hashtags, or specific users. You can also specify the number of tweets to extract using the count parameter.
- **Process and Analyze Tweets**: Process the extracted tweets and perform analysis on

them using various Python libraries such as Pandas, NumPy, or Scikit-learn. • **Visualize Results**: Use data visualization tools such as Matplotlib or Seaborn to create charts and graphs to display the results of your analysis.

- **Store Data**: Save the extracted tweets and any analysis results to a file or database for future use.

Steps to Analyze the Data using NLP:

- Process the cleaned data.
- Use the NRC Word-Emotion Association Lexicon (have aka EmoLex). NRC Word-Emotion Association Lexicon is a list of English with real-valued scores of intensity for eight basic emotion words ns (anger, anticipation, disgust, fear, joy, sadness, surprise, and trust).
- Classify the text content of tweeter dataset created by fetching tweets across the world into basic emotions like anger, anticipation, disgust, fear, joy, sadness, surprise and trust

6. Post-Experiments Exercise

A. Extended Theory:

a. Extract data from one other social media platform and analyze it. Explain the process of extraction and libraries used.

B. Conclusion:

- Write what was performed in the program (s) .
- What is the significance of the program and what Objective is achieved?

C. References:

[1]https://towardsdatascience.com/twitter-sentiment-analysis-classification-using-nltk-python fa9125786l4c
[2]https://www.analyticsvidhya.com/blog/2021/06/covid-19-tweets-analysis-using-nlp-with python/

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Laboratory Exercise

1.Importing Required Libraries

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import re
import nltk
nltk.download('stopwords')
from nltk.corpus import stopwords
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

2. Loading and Displaying the Dataset

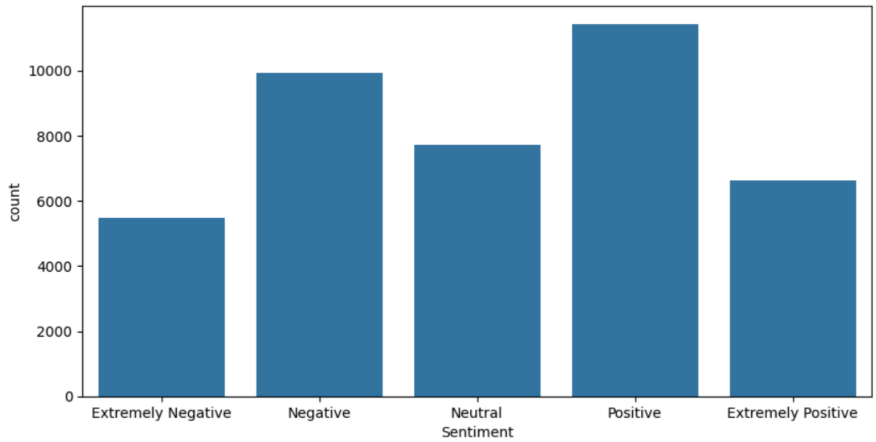
```
data = pd.read_csv("/content/Corona_NLP_train.csv",encoding='latin1')
df = pd.DataFrame(data)
df.head()
```

	UserName	ScreenName	Location	TweetAt	OriginalTweet	Sentiment
0	3799	48751	London	16-03-2020	@MeNyrbie @Phil_Gahan @Chrisitv https://t.co/i...	Neutral
1	3800	48752	UK	16-03-2020	advice Talk to your neighbours family to excha...	Positive
2	3801	48753	Vagabonds	16-03-2020	Coronavirus Australia: Woolworths to give elde...	Positive
3	3802	48754	NaN	16-03-2020	My food stock is not the only one which is emp...	Positive
4	3803	48755	NaN	16-03-2020	Me, ready to go at supermarket during the #COV...	Extremely Negative

3.Visualizing Sentiment Distribution

```
plt.figure(figsize=(10,5))
sns.countplot(x='Sentiment', data=df, order=['Extremely Negative', 'Negative', 'Neutral', 'Positive', 'Extremely Positive'], )
```

<Axes: xlabel='Sentiment', ylabel='count'>




```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 41157 entries, 0 to 41156
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   UserName       41157 non-null  int64
 1   ScreenName     41157 non-null  int64
 2   Location       32567 non-null  object
 3   TweetAt        41157 non-null  object
 4   OriginalTweet  41157 non-null  object
 5   Sentiment      41157 non-null  object
dtypes: int64(2), object(4)
memory usage: 1.9+ MB
```

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4.Cleaning Tweets using Regular Expressions

```
reg = re.compile("(@[A-Za-z0-9]+)|(#[A-Za-z0-9]+)|([^0-9A-Za-z t])|(w+://S+)")
tweet = []
for i in df["OriginalTweet"]:
  tweet.append(reg.sub(" ", i))
df = pd.concat([df, pd.DataFrame(tweet, columns=["CleanedTweet"])], axis=1, sort=False)
```

```
df.head()
```

	UserName	ScreenName	Location	TweetAt	OriginalTweet	Sentiment	CleanedTweet
0	3799	48751	London	16-03-2020	@MeNyrbie @Phil_Gahan @Chrisitv https://t.co/i...	Neutral	Gahan https t co iFz9FAn2Pa and https ...
1	3800	48752	UK	16-03-2020	advice Talk to your neighbours family to excha...	Positive	advice Talk to your neighbours family to excha...
2	3801	48753	Vagabonds	16-03-2020	Coronavirus Australia: Woolworths to give elde...	Positive	Coronavirus Australia Woolworths to give elde...
3	3802	48754	NaN	16-03-2020	My food stock is not the only one which is emp...	Positive	My food stock is not the only one which is emp...
4	3803	48755	NaN	16-03-2020	Me, ready to go at supermarket during the #COV...	Extremely Negative	Me ready to go at supermarket during the ou...

5.TF-IDF Vectorization of Cleaned Tweets

```
from sklearn.feature_extraction.text import TfidfVectorizer
stop_words = set(stopwords.words('english'))     # make a set of stopwords
vectoriser = TfidfVectorizer(stop_words=None)
```

6. Training the Naïve Bayes Model on Tweet Data

```
X_train = vectoriser.fit_transform(df["CleanedTweet"])
# Encoding the classes in numerical values
from sklearn.preprocessing import LabelEncoder
encoder = LabelEncoder()
y_train = encoder.fit_transform(df['Sentiment'])
from sklearn.naive_bayes import MultinomialNB
classifier = MultinomialNB()
classifier.fit(X_train, y_train)
```

```
▾ MultinomialNB
MultinomialNB()
```

7. Loading the Test Dataset for Model Evaluation

```
# importing the Test dataset for prediction and testing purposes
test_data = pd.read_csv("/content/Corona_NLP_test.csv",encoding='latin1')
test_df = pd.DataFrame(test_data)
test_df.head()
```

	UserName	ScreenName	Location	TweetAt	OriginalTweet	Sentiment
0	1	44953	NYC	02-03-2020	TRENDING: New Yorkers encounter empty supermar...	Extremely Negative
1	2	44954	Seattle, WA	02-03-2020	When I couldn't find hand sanitizer at Fred Me...	Positive
2	3	44955	NaN	02-03-2020	Find out how you can protect yourself and love...	Extremely Positive
3	4	44956	Chicagoland	02-03-2020	#Panic buying hits #NewYork City as anxious sh...	Negative
4	5	44957	Melbourne, Victoria	03-03-2020	#toiletpaper #dunnypaper #coronavirus #coronav...	Neutral

8. Cleaning the Test Dataset Tweets

```
reg1 = re.compile("(@[A-Za-z0-9]+)|(#[A-Za-z0-9]+)|([^0-9A-Za-z t])|(w+://S+)")
tweet = []
for i in test_df["OriginalTweet"]:
  tweet.append(reg1.sub(" ", i))
test_df = pd.concat([test_df, pd.DataFrame(tweet, columns=["CleanedTweet"])], axis=1, sort=False)
test_df.head()
```

	UserName	ScreenName	Location	TweetAt	OriginalTweet	Sentiment	CleanedTweet
0	1	44953	NYC	02-03-2020	TRENDING: New Yorkers encounter empty supermar...	Extremely Negative	TRENDING New Yorkers encounter empty supermar...
1	2	44954	Seattle, WA	02-03-2020	When I couldn't find hand sanitizer at Fred Me...	Positive	When I couldn t find hand sanitizer at Fred Me...
2	3	44955	NaN	02-03-2020	Find out how you can protect yourself and love...	Extremely Positive	Find out how you can protect yourself and love...
3	4	44956	Chicagoland	02-03-2020	#Panic buying hits #NewYork City as anxious sh...	Negative	buying hits City as anxious shoppers stock...
4	5	44957	Melbourne, Victoria	03-03-2020	#toiletpaper #dunnypaper #coronavirus #coronav...	Neutral	19 One week everyone...

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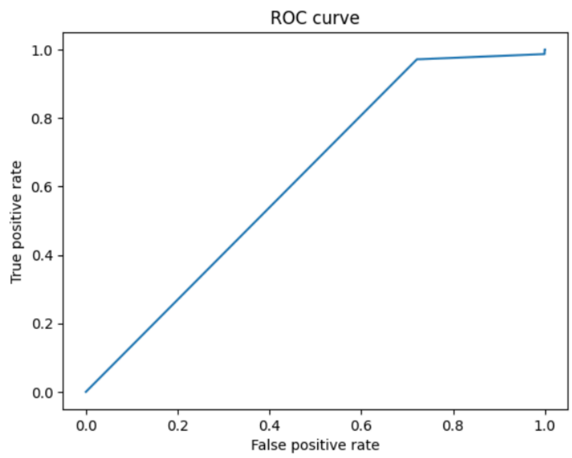
9.Transforming Test Data and Making Predictions

```
X_test = vectoriser.transform(test_df["CleanedTweet"])
y_test = encoder.transform(test_df["Sentiment"])
# Prediction
y_pred = classifier.predict(X_test)
pred_df = pd.DataFrame({'Actual': y_test, 'Predicted': y_pred})
pred_df.head()
```

	Actual	Predicted
0	0	4
1	4	4
2	1	4
3	2	2
4	3	2

10.Evaluating Model Performance using ROC Curve and AUC Score

```
from sklearn import metrics
# Generate the roc curve using scikit-learn.
fpr, tpr, thresholds = metrics.roc_curve(y_test, y_pred, pos_label=1)
plt.plot(fpr, tpr)
plt.xlabel('False positive rate')
plt.ylabel('True positive rate')
plt.title('ROC curve')
plt.show()
# Measure the area under the curve. The closer to 1, the "better" the predictions.
print("AUC of the predictions: {0}".format(metrics.auc(fpr, tpr)))
```




```
AUC of the predictions: 0.6231713165790018
```

---

6. Post-Experiments Exercise

A. Extended Theory:

a. Extract data from one other social media platform and analyze it. Explain the process of extraction and libraries used.

Dataset Used: Reddit Mental Health Dataset (RMHD) (2022)

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import os
import pandas as pd
import numpy as np
import re
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
import nltk
from nltk.corpus import stopwords
from textblob import TextBlob

nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
True
```

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```
base_dir = "/content/drive/MyDrive/AIML in Healthcare/7/2022"
```

```
def load_reddit_data(base_dir):
    all_data = []
    for month_folder in os.listdir(base_dir):
        month_path = os.path.join(base_dir, month_folder)
        if os.path.isdir(month_path):
            for file in os.listdir(month_path):
                if file.endswith(".csv"):
                    file_path = os.path.join(month_path, file)
                    try:
                        df = pd.read_csv(file_path)
                        df['Month'] = month_folder
                        # topic inferred from filename (e.g., anxapr22.csv → anxiety)
                        topic = re.findall(r'[a-zA-Z]+', file)[0]
                        df['Topic'] = topic.lower()
                        all_data.append(df)
                    except Exception as e:
                        print(f"⚠️ Could not load {file}: {e}")
    combined_df = pd.concat(all_data, ignore_index=True)
    return combined_df

reddit_df = load_reddit_data(base_dir)
print("✅ Data Loaded Successfully!")
print(f"Total Records: {len(reddit_df)}")
print(reddit_df.head())
```

```
✅ Data Loaded Successfully!
Total Records: 389387
   Unnamed: 0                 author  created_utc  score  \
0           0          Hollow_space00   1651325610      1
1           1  throwawaycauseisad9   1651325490      1
2           2              [deleted]   1651325489      1
3           3          Pomeranian111   1651325382      1
4           4  InternationalAd7273   1651325307      1

                                            selftext  \
0  You're grown up live a life. That's what I kee...
1  Only my closest family would care about my dis...
2                                         depression
3  I'm also an ugly 23-year-old virgin, turning 2...
4  I know, it sounds absolutely insane and I feel...

                     subreddit  \
0                   depression
1                   depression
2  How do you deal with depression?
3                   depression
4                   depression

                                               title            timestamp  \
0           Another tiring day for this tiring soul.  2022-04-30 23:33:30
1          I'm just a side quest in everyone's story  2022-04-30 23:31:30
2  https://www.reddit.com/r/depression/comments/u...  2022-04-30 23:31:29
3  Being lonely and depressed is a horrible combi...  2022-04-30 23:29:42
4                           I miss my depression.  2022-04-30 23:28:27

    Month   Topic
0  Apr 22  depapr
1  Apr 22  depapr
2  Apr 22  depapr
3  Apr 22  depapr
4  Apr 22  depapr
```

```
text_col = None
for col in ['body', 'selftext', 'text', 'content', 'clean_text']:
    if col in reddit_df.columns:
        text_col = col
        break

if text_col is None:
    text_col = reddit_df.columns[0]  # fallback

print(f"📝 Using text column: {text_col}")
```

```
📝 Using text column: selftext
```

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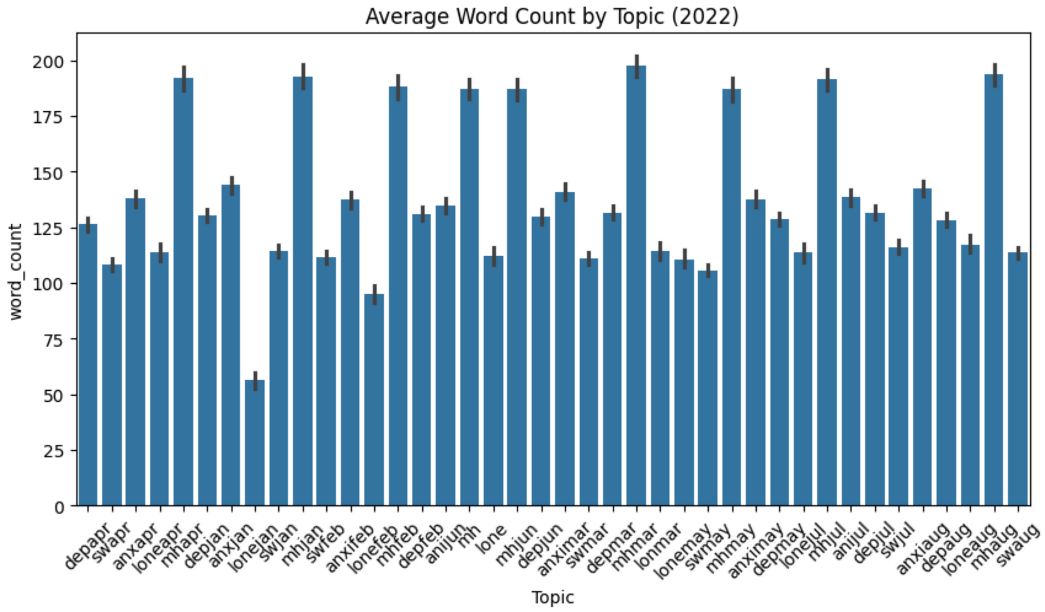
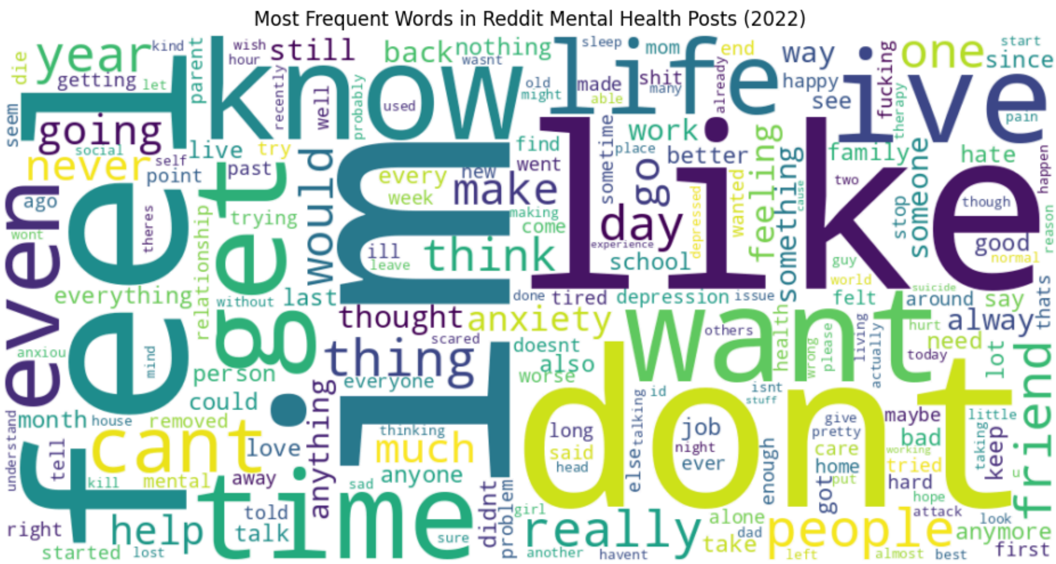
```python
def clean_text(text):
    text = str(text).lower()
    text = re.sub(r'http\S+', '', text)        # remove URLs
    text = re.sub(r'[^a-z\s]', '', text)       # remove punctuation/numbers
    text = re.sub(r'\s+', ' ', text)           # remove extra spaces
    return text.strip()

reddit_df['clean_text'] = reddit_df[text_col].apply(clean_text)
```

```python
reddit_df['word_count'] = reddit_df['clean_text'].apply(lambda x: len(x.split()))

plt.figure(figsize=(10,5))
sns.barplot(x='Topic', y='word_count', data=reddit_df)
plt.title('Average Word Count by Topic (2022)')
plt.xticks(rotation=45)
plt.show()
```




```python
stop_words = set(stopwords.words('english'))
text_all = " ".join(reddit_df['clean_text'])
wordcloud = WordCloud(width=1000, height=500, background_color='white',
                      stopwords=stop_words, collocations=False).generate(text_all)
plt.figure(figsize=(12,6))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("Most Frequent Words in Reddit Mental Health Posts (2022)")
plt.show()
```



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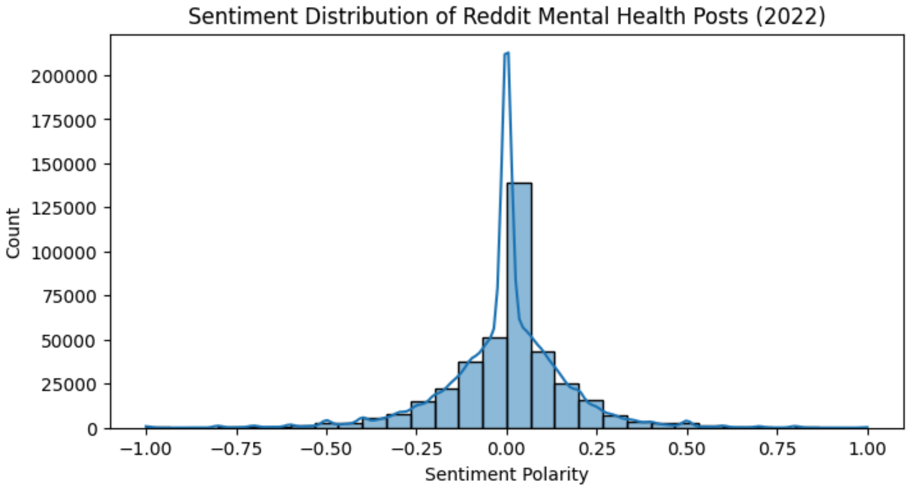
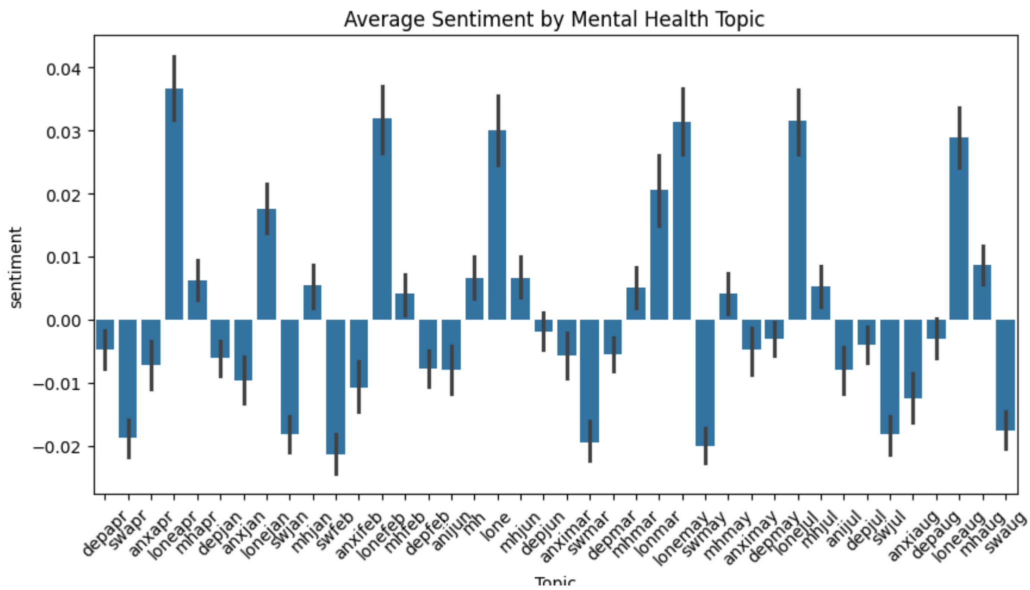
```python
def get_sentiment(text):
    return TextBlob(text).sentiment.polarity

reddit_df['sentiment'] = reddit_df['clean_text'].apply(get_sentiment)

# Sentiment distribution
plt.figure(figsize=(8,4))
sns.histplot(reddit_df['sentiment'], bins=30, kde=True)
plt.title("Sentiment Distribution of Reddit Mental Health Posts (2022)")
plt.xlabel("Sentiment Polarity")
plt.ylabel("Count")
plt.show()

# Average sentiment by topic
plt.figure(figsize=(10,5))
sns.barplot(x='Topic', y='sentiment', data=reddit_df)
plt.title("Average Sentiment by Mental Health Topic")
plt.xticks(rotation=45)
plt.show()
```





Explanation:

<u>Libraries Used and Their Purpose</u>

1. os → Navigate folders and file paths.
2. pandas → Read CSVs, store and manipulate data.
3. numpy → Numerical operations (optional here).
4. re → Text cleaning with regular expressions.
5. matplotlib & seaborn → Data visualization (bar plots, histograms).
6. wordcloud → Generate word clouds to visualize frequent words.
7. nltk → Provides stopwords for text preprocessing.
8. textblob → Performs sentiment analysis (polarity computation).

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Process of Data Extraction & Analysis

The data extraction and analysis process involved loading all monthly Reddit CSV files, reading each file into a DataFrame, and adding Month and Topic columns to provide contextual information. All individual DataFrames were then concatenated into a single large dataset. The text was cleaned by converting it to lowercase, removing URLs, punctuation, and numbers, eliminating extra spaces, and selecting the main text column, either body or the first column. Basic text analysis was performed by counting the number of words in each post and visualizing the average word count by topic. A word cloud was generated by combining all cleaned text, removing common stopwords, and creating a visual representation of the most frequent words. Additionally, sentiment analysis was conducted using TextBlob to compute sentiment polarity for each post, and the results were visualized through a histogram of sentiment distribution and a bar plot showing the average sentiment by topic.